

Vrunda Joshi

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SUMMARY

Mathematics graduate pursuing an M.Sc. in Data Science at Dhirubhai Ambani University, with a strong foundation in statistics, probability, and analytical problem-solving. Skilled in Python, SQL, and machine learning through academic projects and research at SARAL Lab; seeking an entry-level Data Analyst / Data Science role.

EXPERIENCE

SARAL Lab (Systems, Autonomy, Robotics & Automation Lab), DAU

May 2026 - Present

Student Researcher

Gandhinagar, Gujarat

- Extracted trip logs from the lab's physics-based simulator of a working coal mine, where self-driving haul trucks navigate a road network of about 240 intersections with every second of every trip recorded.
- Visualized raw traffic patterns and preprocessed the trip data: cleaning, structuring, and preparing features to make it modeling-ready.
- Trained and benchmarked trip-duration models; the best (Random Forest) was off by just 16 seconds on average, about 7% of a typical trip, against baselines of 64 seconds (linear regression) and 18 seconds (gradient boosting).
- Analyzed and visualized the dispatch-policy experiments: four Double DQN (RL) policies against the hand-built rule baseline (ETA_MIN) over 5 seeds and 100k simulated trips; all four beat ETA_MIN on reward (+5.2% to +10.7%), the best per-mine policy (-267.4) beat even a greedy oracle (-272.3), with 13.5% lower queue share.
- Built a live operations dashboard that turns model outputs into a monitoring view of fleet KPIs and forecasted congestion hotspots on the mine road graph, supporting dispatch decisions.

PROJECTS

Screen Time & Sleep: Multi-Scale Statistical Study | *Python (statsmodels), SQL (SQLite)* |

Winter 2026

- Designed a two-scale study of whether bedtime screen use predicts sleep duration: a self-conducted stratified pilot (n = 30 students, paired exam and regular weeks) plus an independent 996-student public dataset (Mendeley Data, CC BY 4.0).
- Analyzed with Spearman and Kendall correlations, HC3-robust OLS, and ordinal-logistic checks, validated by residual, Q-Q, and Cook's Distance diagnostics plus an MCAR/MAR/MNAR sensitivity simulation; built a reproducible pipeline with a SQLite SQL layer (GROUP BY, CTEs, window functions), pytest on GitHub Actions CI, and pinned dependencies.
- Both scales rejected the claim that screens cost sleep: pilot exam-week slope $p = 0.58$ with the 1.15 h exam-week sleep loss pinned on workload, and the 996-student association (+0.38 h per use level) halved to +0.19 h after caffeine, stress, and activity adjustment, consistent with schedule-flexibility confounding via a causal DAG; pre-registered $n \geq 100$ follow-up (87% power, phone-log capture) in collection.

Heart Disease Prevalence: Interactive Data Visualization | *scikit-learn, SHAP, Tableau* |

Autumn 2026

- Analyzed a public clinical dataset (Kaggle/UCI); audit found 723 exact duplicate patient records in the raw 1,025-row file and removed them (302 unique patients) to prevent train/test leakage, then trained a Random Forest for feature importance, validated on a stratified 61-patient held-out test set (accuracy 0.787, ROC-AUC 0.865).
- Ranked feature importance three ways (impurity-based, permutation on held-out data, SHAP for direction), then separately within male and female subgroups with 500-iteration bootstrap intervals; the ranking shifts descriptively by gender (men: vessel count and ST depression; women: exercise-induced angina and thalassemia test), reported as suggestive because every interval overlaps at $n = 302$.
- Built an interactive Tableau dashboard (gender-split importance, prevalence heatmap, grouped bars; Gender, Age group, and Chest-pain type filters), presented to the class with the course instructor as project jury.

TECHNICAL SKILLS

Languages: Python, SQL

Libraries & Tools: pandas, NumPy, scikit-learn, statsmodels, Matplotlib, Tableau, Excel, Git/GitHub

Concepts: Statistical Inference, Hypothesis Testing, Regression Modeling, Data Cleaning & Visualization

EDUCATION

Dhirubhai Ambani University

2025 - Present

M.Sc. Data Science

Gandhinagar, Gujarat

Sir P.T. Sarvajani College of Science

2022 - 2025

B.Sc. in Mathematics

Surat, Gujarat